

TECHNICAL REPORT

IEC
61010-3-061

First edition
1999-02

**Safety requirements for electrical equipment
for measurement, control, and laboratory use –**

**Part 3-061:
Conformity verification report for IEC 61010-2-061,
Particular requirements for laboratory atomic
spectrometers with thermal atomization and ionization**

*Règles de sécurité pour appareils électriques de mesure,
de régulation et de laboratoire –*

*Partie 3-061:
Rapport de vérification de la conformité de la CEI 61010-2-061,
Spécifications particulières pour les spectromètres
à atomisation et ionisation thermiques*



Reference number
IEC 61010-3-061:1999(E)

Numbering

As from 1 January 1997 all IEC publications are issued with a designation in the 60000 series.

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For graphical symbols, and letter symbols and signs approved by the IEC for general use, readers are referred to publications IEC 60027: *Letter symbols to be used in electrical technology*, IEC 60417: *Graphical symbols for use on equipment. Index, survey and compilation of the single sheets* and IEC 60617: *Graphical symbols for diagrams*.

* See web site address on title page.

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Commission Electrotechnique Internationale
International Electrotechnical Commission
Международная Электротехническая Комиссия

PRICE CODE

XB

For price, see current catalogue

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**SAFETY REQUIREMENTS FOR ELECTRICAL EQUIPMENT
FOR MEASUREMENT, CONTROL, AND LABORATORY USE –****Part 3-061: Conformity verification report for IEC 61010-2-061,
Particular requirements for laboratory atomic spectrometers
with thermal atomization and ionization**

FOREWORD

- 1) The IEC (International Electrotechnical Commission) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of the IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, the IEC publishes International Standards. Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. The IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
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Technical reports do not necessarily have to be reviewed until the data they provide are considered to be no longer valid or useful.

IEC 61010-3-061, which is a technical report, has been prepared by IEC technical committee 66: Safety of measuring, control, and laboratory equipment.

The text of this technical report is based on the following documents:

Enquiry draft	Report on voting
66/182/CDV	66/209/RVC

Full information on the voting for the approval of this technical report can be found in the report on voting indicated in the above table.

This report is a Technical Report and is of a purely informative nature and is therefore by itself not to be regarded as an international Standard. It is for use by testhouses and other users to assist them with determining and recording verification of conformity of the equipment under test with the requirements of:

IEC 61010-2-061:1995
and
IEC 61010-1:1990 + amendment 1:1992,
or
IEC 61010-1:1990 + amendment 1:1992 + amendment 2:1995

The protocol for completion of this report is contained in publication IEC 61010-3: 1997.

Requirements which apply only to amendment 2 to IEC 61010-1 are indicated by "(AM 2)" in the first column. Requirements deleted by amendment 2 to IEC 61010-1 are indicated by "(AM 1 only)" in the second column.

The IEC sells read-only PDF files as a general rule. In the present instance, and quite exceptionally, to enable the user to fill in the forms, a revisable file is included in a pocket affixed to the back cover of this publication.

This publication can be down-loaded from the Web as a PDF file. There is, however, at the end of the document, a revisable file containing the forms. Please use the zip/unzip function.

Conformity Verification Report IEC 61010-2-061: Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 2-061:1995, Particular requirements for laboratory atomic spectrometers with thermal atomization and ionization	
Report reference No: Compiled by (+ signature).....: Approved by (+ signature): Date of issue.....:	
Testing organization: Address.....: Testing location:	
Applicant.....: Address.....:	
Standard:	IEC 61010-2-061:1995 and IEC 61010-1:1990 + A1:1992 or IEC 61010-1:1990 + A1:1992 + A2:1995
Copyright blank test report:	This report has been prepared by IEC TC 66, which retains responsibility for any changes or corrections required.
Test procedure: Procedure deviation: Non-standard test method:	
Type of item tested: Laboratory Trade mark.....: Model/type reference: Manufacturer: Rating:	
Copy of rating plate:	

Description of equipment function

INSTALLATION/OVERVOLTAGE CATEGORY

POLLUTION DEGREE:

Environmental rating: Standard Other (specify):

Equipment mobility: Portable Hand-held Floorstanding Fixed
 Built in Benchmounted Other (specify):

Connection to mains supply: Permanent Detachable Non detachable None

Operating conditions: Continuous Short-time Intermittent

Overall size of the equipment (length × width × height):

Mass of the equipment (kg):

Marked degree of protection to IEC 60529: IP__ __

Accessories and detachable parts included in the evaluation:

Options:

NOTE – "(see Form A.X)" refers to a Form appended to the report.

Table 1 – Documents attached to this report

Document No.	Document description	Number of pages

Table 2 – Test equipment list

[illegible]

Table 3 – List of components relied on for safety

Unique component reference or location (including drawing reference if required)	Application/Function	Manufacturer and part number (note 1)	RATING (note 2)	Licence number, file number or other documentary evidence of acceptance
NOTE 1 – List all manufacturers concerned. NOTE 2 – Electrical, mechanical, flammability, etc.				

Clause Subclause	Requirement	Result	Comments
5	Marking and documentation		
5.1.1	General Required equipment markings are: a) visible: – from the exterior or – after removing a cover or – opening a door or – after removal from a rack or panel b) not put on parts which can be removed by an OPERATOR c) Letter symbols (IEC 60027) used d) Graphic symbols (IEC 61010-1: table 1) used		
5.1.2	Identification Equipment is identified by: – manufacturer's name or registered trade mark – model number, name or other means – the degree of protection, if any, according to IEC 60529		
5.1.3	Mains supply Equipment is marked as follows: a) nature of supply: – a.c. RATED mains frequency or range of frequencies – d.c. with symbol 1 b) RATED supply voltage(s) or range c) – maximum RATED power (W or VA) or input current		
(AM 2)	If more than one voltage range: – separate values marked or – values differ by less than 20 % (see Form A.3)		
F			

Clause Subclause	Requirement	Result	Comments
5.1.4	d) OPERATOR – set for different RATED supply voltages:		
	– indicates the equipment set voltage		
	– PORTABLE EQUIPMENT indication is visible from the exterior		
	– changing the setting changes the indication		
	e) Accessory mains socket-outlets accepting standard mains plugs are marked:		
	– with the voltage if it is different from the mains supply voltage		
	– for use only with specific equipment		
	If not marked for specific equipment it is marked with:		
	– the maximum RATED current or power, and maximum permitted leakage current		
	or		
	– symbol 14 with full details in the documentation		
	F The measured value not more than 110 % (see Form A.3)		
	Fuses		
	OPERATOR replaceable fuse marking (see also 5.4.5)		
	5.1.5 Measuring circuit TERMINALS		
	(AM 2) RATED maximum working voltage or current marked		
	Unless clear indication that below limits:		
	– maximum RATED voltage to earth is marked		
	or		
	– for specific connection only, and means for identifying provided		
	– is adjacent to TERMINALS		
	or		
	– if insufficient space:		
	– on the RATING plate or scale plate		
	or		
	– if the TERMINAL is marked with symbol 14		

Clause Subclause	Requirement	Result	Comments
5.1.6	(AM 2) INSTALLATION CATEGORY marked		
	(AM 2) TERMINALS permanently connected and not ACCESSIBLE		
	TERMINALS and operating devices		
	Where necessary for safety, indication of purpose of TERMINALS, connectors, controls and indicators		
	(AM 2) Mains supply TERMINALS identified		
	(AM 2) Power supply switch on or off position marked if used as disconnecting device		
	TERMINAL marking:		
	a) FUNCTIONAL EARTH TERMINALS		
	b) PROTECTIVE CONDUCTOR TERMINALS:		
	– symbol 6 is placed close to or on the TERMINAL		
	or		
	– part of appliance inlet		
	c) TERMINALS of measuring and control circuits		
	d) TERMINALS supplied from the interior		
	e) ACCESSIBLE FUNCTIONAL EARTH TERMINALS		
	f) Position on or off of power supply switch (AM 1 only)		
5.1.7	Equipment protected by DOUBLE INSULATION or REINFORCED INSULATION		
	Protected throughout (symbol 11 used)		
	Only partially protected (symbol 11 not used)		
5.1.8	Battery charging		
	Equipment with means to charge rechargeable batteries is marked:		
	– to warn against the charging of non-rechargeable batteries		
	– to indicate the type of rechargeable battery used		

Clause Subclause	Requirement	Result	Comments
5.1.101	Gas and liquid connections Connectors unambiguously marked with: – identity of gas or liquid and – maximum pressure		
5.2	Warning markings – visible when ready for NORMAL USE – if necessary marked with symbol 14 – are near or on applicable parts (AM 2) – statement to isolate or disconnect – advice how to avoid contact with ACCESSIBLE HAZARDOUS LIVE parts – TERMINAL voltage exceeding 1 kV (symbol 12) – easily touched high temperature parts (symbol 13) – warning for more than 2 s (AM 1 only) Protective means against hot gases or plasma marked with symbol 13		
5.3	Durability of markings F The required markings remain clear and legible in NORMAL USE (see Form A.4)		
5.4	Documentation		
5.4.1	General Equipment is accompanied by documentation which includes: – technical specification – instructions for use – name and address of manufacturer or supplier (AM 2) Definition of INSTALLATION CATEGORY Warning statements and a clear explanation of warning symbols: – provided in the documentation or – information is marked on the equipment		

Clause Subclause	Requirement	Result	Comments
5.4.2	Equipment RATINGS		
	Documentation includes:		
	– supply voltage or voltage range		
	– frequency or frequency range		
	– power or current RATING		
	– a description of all input and output connections		
	– RATING of insulation of external circuits, when such circuits are nowhere ACCESSIBLE		
	– statement of the range of environmental conditions		
5.4.3	Equipment installation		
	Documentation includes instructions for:		
	– assembly, location and mounting		
	– protective earthing		
	– connections to supply		
	– ventilation requirements		
	– requirements for supply connection of air, fuel gases, etc.		
	– user shall ensure that the type of connector used at the outlet side of the gas pressure regulator conforms to national requirements		
	– requirement for liquid connection		
	– requirement for fume extraction of hazardous gases		
	– requirements for trap of hazardous sample residues in the exhaust gas stream:		
	– appropriate filter		
	or		
	– other system		
	– instruction that user carries out appropriate leakage tests		
	– instruction for examining parts of the external gas supply system		
	– instruction for provisions for collecting waste of spray-chamber		

Clause Subclause	Requirement	Result	Comments
5.4.4	(AM 2) – maximum sound power level		
	(AM 2) – instructions about sound pressure		
	Additional for permanently connected equipment:		
	– supply wiring		
	– any external switch or circuit-breaker (including location)		
	– any external overcurrent protection		
	Equipment operation		
	Instructions for use include:		
	– identification of operating controls		
	(AM 2) – positioning for disconnection		
	– interconnection		
	– specification of intermittent operation limits		
	– explanation of symbols used		
	– replacement of consumable materials		
	– cleaning and decontamination (see 11.2)		
	– a reminder about correct collection and disposal of waste materials, including:		
	a) a suitably sized waste container		
	b) provisions for the removal into an exhaust system of any gases		
	– list of fluids known to be potentially unsafe if used with the equipment		
	– statement that the list is not complete; contact the manufacturer		
	A warning against use in a manner not specified by the manufacturer		
	5.4.101 Cleaning and decontamination		
	Documentation indicates:		
	– user responsibility after hazardous material spillage		

Clause Subclause	Requirement	Result	Comments
5.4.5	– recommendations for cleaning/ decontamination and generic names		
	Wording "Before using any cleaning or, etc." in documentation		
	If decontamination by steam, withstands one method of table 101		
	Equipment maintenance		
	Instructions include:		
	– sufficient preventive maintenance and inspection information		
	(AM 2) – replacement of hoses, etc.		
	– specific battery type		
	– any manufacturer specified parts		
	– RATING and characteristics of fuses		
6	Protection against electric shock		
	F (see Form A.5)		
6.1	General		
	Conformity is checked by the determination of 6.2 and 6.3 followed by the tests of 6.4 to 6.12		
6.1.1	Exceptions		
	F Capacitance test (see Forms A.6 and A.7)		
6.2	Determination of ACCESSIBLE parts		
	F (see Form A.6)		
6.3	Permissible limits for ACCESSIBLE parts		
6.3.1	Values in NORMAL CONDITION		
	F (see Form A.7)		
6.3.2	Values in SINGLE FAULT CONDITION		
	F (see Form A.8)		
6.4	Protection in NORMAL CONDITION (see 6.8 and 8.1)		
6.5	Protection in SINGLE FAULT CONDITION		
	Additional protection is provided by:		
	– one or more of 6.5.1 to 6.5.3		
	or		
	– automatic disconnection of the supply		

Clause Subclause	Requirement	Result	Comments
6.5.1	Protective earthing ACCESSIBLE conductive parts: – bonded to the protective conductor terminal or – separated by screen or BARRIER from parts which are HAZARDOUS LIVE (For indirect bonding of measurement and test equipment see 6.5.1.4)		
6.5.1.1	PROTECTIVE BONDING PROTECTIVE BONDING consists of directly connected structural parts or discrete conductors or both; and withstands thermal and dynamic stresses		
6.5.1.2	F Bonding impedance of plug-connected equipment (see Form A.9)		
6.5.1.3	F Bonding impedance of PERMANENTLY CONNECTED EQUIPMENT (see Form A.9)		
6.5.1.4	F Indirect bonding for measuring and test equipment (see Form A.9)		
6.5.2	DOUBLE INSULATION and REINFORCED INSULATION (see 6.7, 6.8 and 6.9.2)		
6.5.3	F PROTECTIVE IMPEDANCE (see Form A.10) Components wires and connections are RATED as required		
6.5.4 or	Built-in equipment (AM 1 only)		
(AM 2)	Built-in panel meters If after building-in the requirements of 6.5.1 to 6.5.3 are not met: – equipment (AM 1 only): or – panel meter: – has no ACCESSIBLE conductive parts – has BASIC INSULATION of ACCESSIBLE surfaces – has DOUBLE/REINFORCED INSULATION of ACCESSIBLE surface of parts intended to be grasped		

Clause Subclause	Requirement	Result	Comments
6.6	External circuits		
6.6.1	Separation of internal circuits (see list in Forms A.1 and A.5)		
F	If the other internal circuit exceeds the values of 6.3.2 in NORMAL CONDITION and only BASIC INSULATION:		
	a) short circuit could not make external circuit HAZARDOUS LIVE		
	b) manufacturer's instructions include:		
	– a statement that the TERMINAL for external circuits is for use only with equipment which has no live parts which are ACCESSIBLE		
	– the RATING of the insulation required for external circuits		
	– the connection to be used at the remote end of external circuits		
	– the type of equipment which may be connected to the TERMINAL		
6.6.2	TERMINALS for external circuits		
	ACCESSIBLE TERMINALS are not HAZARDOUS LIVE except as permitted by 6.1.1		
	The following TERMINALS are not HAZARDOUS LIVE:		
	– PROTECTIVE CONDUCTOR TERMINALS		
	– FUNCTIONAL EARTH TERMINALS		
	– headphone TERMINALS		
F	TERMINALS which receive a charge from an internal capacitor (see Form A.7)		
	High-voltage TERMINALS energized from the interior are:		
	– not accessible		
	or		
(AM 2)	– marked		

Clause Subclause	Requirement	Result	Comments
6.6.3	Circuits with TERMINALS which are HAZARDOUS LIVE These circuits: – are not connected to ACCESSIBLE conductive parts or – are connected to ACCESSIBLE conductive parts, but are not mains circuits and have one TERMINAL contact at earth potential No ACCESSIBLE conductive parts are HAZARDOUS LIVE		
6.7	F CLEARANCES and CREEPAGE DISTANCES (See annex D of IEC 61010-1 and Form A.11)		
6.8	F Dielectric strength tests (see annex E of IEC 61010-1 and Form A.12) Protection against the spread of fire (see 9.1)		
6.9	Constructional requirements for protection against electric shock		
6.9.1	General In circuits exceeding the values of 6.3.2: – security of wiring connections – screws securing removable covers – accidental loosening		
6.9.2	ENCLOSURES of equipment with DOUBLE INSULATION OR REINFORCED INSULATION ENCLOSURE surrounds all metal parts except for small metal parts which are separated ENCLOSURES or parts made of insulating material Protection for metal ENCLOSURES or parts by: – PROTECTIVE IMPEDANCE or – an insulating coating or BARRIER on the inside or – CLEARANCES and CREEPAGE DISTANCES cannot be reduced by loosening of parts or wires		

Clause Subclause	Requirement	Result	Comments
6.9.3	Equipment using PROTECTIVE BONDING		
	a) OPERATOR removable parts		
	b) Movable conductive connections		
	c) Exterior metal braids of cables		
	d) Mains passed through the equipment		
	e) Protective earthing conductors green/yellow		
	Exceptions:		
	– earthing braids		
	– internal protective conductors		
	f) Equipment using PROTECTIVE BONDING		
6.9.4 (AM 2)	Over-range indication		
	Unambiguous		
6.10	Connection to mains supply source (AM 1 only)		
or			
(AM 2)	Connection to mains supply source and connections between parts of equipment		
6.10.1	Mains supply cords		
	RATED for maximum equipment current (see 5.1.3c)		
	Cable complies with IEC 60227 or IEC 60245 or is a certified cord		
(AM 2)	Heat resistant if likely to contact hot parts		
(AM 2)	Temperature RATING (cord and inlet)		
	Green/yellow used only for connection to PROTECTIVE CONDUCTOR TERMINALS		
	Combination of conductors (AM 1 only):		
	– all have same degree of insulation		
	or		
	– no hazard from short circuits		

Clause Subclause	Requirement	Result	Comments
6.10.2	Detachable cords with IEC 60320 mains connectors:		
	– comply with IEC 60799		
	or		
	– have the current RATING of the mains connector		
	Fitting of non-detachable mains supply cords		
	6.10.2.1 Cord entry		
	Non-detachable cord protection:		
	– inlet smoothly rounded with radius $\geq 1,5 D$		
	or		
	– insulated cord guard protruding $\geq 5 D$		
	6.10.2.2 Cord anchorage		
	The protective earth conductor is the last to take the strain		
	Cord anchorages:		
	– the cord is not clamped by direct pressure from a screw		
	– knots are not used		
6.10.2.2	– cannot push the cord into the equipment to cause a hazard		
	– no failure of cord insulation in anchorage with metal parts		
	– compression bushing:		
	a) clamps all types and sizes of mains cords and		
	b) is suitable:		
	i) for connection to TERMINALS provided		
	or		
	ii) it is designed for screened mains cord		
	– cord replacement does not cause a hazard and method of strain relief is clear		
	F Push-pull test (see Form A.13)		

Clause Subclause	Requirement	Result	Comments
6.10.3	Plugs and connectors		
	a) Mains supply plugs, connectors etc., conform with relevant specifications		
	b) If equipment supplied at voltages below 6.3.2.1:		
	– plugs of mains supply cords do not fit mains sockets above RATED supply voltage		
	– mains type plugs used only for connection to mains supply		
	c) Plug pins which receive a charge from an internal capacitor (see Form A.7)		
	d) Accessory mains socket outlets:		
	– if a standard mains plug is accepted, there is a marking (see 5.1.3e)		
	– input has a protective earth conductor if outlet has earth TERMINAL contact		
6.11	TERMINALS		
6.11.1	ACCESSIBLE TERMINALS		
	a) No risk of accidental contact (see also 5.1.6c)		
	b) Will not work loose		
6.11.2	PROTECTIVE CONDUCTOR TERMINAL		
	a) Appliance inlet (no requirement)		
	b) For rewirable cords and PERMANENTLY CONNECTED EQUIPMENT, PROTECTIVE CONDUCTOR TERMINAL is close to mains supply TERMINALS		
	c) If no mains supply is required, any PROTECTIVE CONDUCTOR TERMINAL:		
	– is near TERMINALS of circuit for which protective earthing is necessary		
	– external if other TERMINALS external		
	d) Equivalent current-carrying capacity to mains supply TERMINALS		

Clause Subclause	Requirement	Result	Comments
(AM 2)	e) – Soldered connections:		
	i) independently secured		
	ii) not used for other purposes		
	– Screw connections are secured		
	f) Contact surfaces are metal		
	g) If plug-in, makes first and breaks last		
	h) Protective conductor of measuring circuit:		
	– current RATING		
	– PROTECTIVE BONDING:		
	i) not interrupted		
	or		
	ii) indirect bonding		
	6.11.3 FUNCTIONAL EARTH TERMINALS		
	Independent connection		
6.12	Disconnection from supply source		
6.12.1	General		
	Disconnection device provided		
6.12.1.1	Exception to 6.12.1		
	Short circuit or overload cannot cause a hazard		
6.12.2	Requirements according to type of equipment		
6.12.2.1	PERMANENTLY CONNECTED EQUIPMENT		
	– switch or circuit-breaker is part of the equipment		
	or		
	– documentation specifies switch location and marking		
6.12.2.2	Single-phase cord-connected equipment		
	– switch or circuit-breakers		
	or		
	– appliance coupler (disconnectable without TOOL)		
	or		
	– separable plug (without locking device)		

Clause Subclause	Requirement	Result	Comments
6.12.2.3	Hazards arising from function		
	Emergency switch		
	Emergency switch ≤1 m from the moving part		
6.12.3	Disconnecting devices		
	Electrically close to the supply		
6.12.3.1	Switches and circuit-breakers		
	When used as disconnection device:		
	– meet IEC 60947-1 and IEC 60947-3		
	– contact separation		
	– contact position evident in off position		
	– marked to indication function		
	– not incorporated in mains cord		
	– does not interrupt protection earth conductor		
	– if has other contacts, meets separation requirements of 6.6 and 6.7		
6.12.3.2	Appliance couplers and plugs		
	Where an appliance coupler or separable plug is used as the disconnecting device (see 6.12.2.2):		
	– readily identifiable and easily reached by the OPERATOR		
	– single-phase PORTABLE EQUIPMENT cord length ≤3 m		
	Protective earth conductor connected first and disconnected last		
7	Protection against mechanical hazards		
7.1	General		
	Conformity is checked by 7.2 to 7.5		
7.2	Moving parts		
	Moving parts not able to crush, etc. (see also 6.12.2.3)		

Clause Subclause	Requirement	Result	Comments
7.3	(AM 2) If OPERATOR access permitted:		
	a) access requires TOOL		
	b) statement about training		
	c) warning markings or symbol 14		
	Stability		
	Marking of non-automatic means		
	Conformity tests:		
	– 10° tilt test		
	– multi-directional force test		
	– downward force test		
7.4	Provisions for lifting and carrying		
	Handles or grips withstand four times mass		
	Equipment ≥18 kg:		
	– has means for lifting or carrying		
	or		
7.5	Expelled parts		
	Equipment contains or limits the energy		
	Protection not removable without the aid of a TOOL		
	Mechanical resistance to shock, vibration and impact (AM 1 only)		
	(AM 2) Mechanical resistance to shock and impact		
8 or	After the tests of 8.1 to 8.4		
	F (see Form A.11):		
	F – voltage tests (see Form A.12)		
	– inspection, equipment meets the following requirements:		
	a) HAZARDOUS LIVE parts not ACCESSIBLE		
	b) ENCLOSURE shows no cracks (hazard)		
	c) CLEARANCES not less than their permitted values (see Form A.11)		
	F		

Clause Subclause	Requirement	Result	Comments
9	– BARRIERS not damaged or loosened		
	– no moving parts exposed, except as permitted by 7.2		
	– no damage which could cause spread of fire		
	Equipment temperature limits and protection against the spread of fire		
	9.1 General		
	Conformity is checked by:		
	– 9.2 and fault tests of 4.4 (see Forms A.1, A.2 and A.18)		
	F or		
	– measurement of CREEPAGE DISTANCE and CLEARANCE and the voltage tests of annex G (see Form A.14)		
	F or		
	– method of annex F (see Forms A.15, A.16 and A.17)		
	F		
	9.2 Temperature tests		
	9.3 Guards		
	F Surfaces liable to exceed 100 °C (see Form A.18):		
	– protected by guards		
	or		
	– marked		
	or		
	– intended to be hot (see 9.1)		
	Guards not removable without TOOL		
	Protection against unintentional contact		
	9.4 Field-wiring TERMINAL boxes		
	Temperature RATING of the cable is:		
	F – marked (see Form A.18)		
	and		
	– adjacent to field-wiring TERMINALS		
	or		
	– visible during and after installation		

Clause Subclause	Requirement	Result	Comments
9.5	Overtemperature protection devices		
F	– fitted, to operate in SINGLE FAULT CONDITION (see Form A.1)		
	– meets 14.3		
	– does not operate in NORMAL USE (see 3.5.6)		
	– if self-resetting, can only be set to operate in SINGLE FAULT CONDITION		
9.6	Overcurrent protection		
9.6.1	PERMANENTLY CONNECTED EQUIPMENT		
	Device:		
	– fitted within the equipment		
	or		
	– specified in manufacturer's instructions		
9.6.2	Other equipment		
	Protection within the equipment		
	Devices not in the protective conductor		
	Fuses or single-pole circuit-breakers not fitted in neutral (multi-phase)		
10	Resistance to heat		
10.1	Integrity of CLEARANCE and CREEPAGE DISTANCES (see Form A.11)		
F			
10.2	Resistance to heat of non-metallic ENCLOSURE (see Form A.19)		
F			
10.3	Resistance to heat of insulation material		
	Parts supporting:		
	– parts connected to mains supply		
	– TERMINALS carrying >0,5 A		
11	Resistance to moisture and liquids (AM 1 only)		
or			
(AM 2)	Protection against hazards from fluids		
11.1	General		

Clause Subclause	Requirement	Result	Comments
11.1.101	Sampling probes Sample fluid on outside of sampling probe shall not cause a hazard		
11.1.101	Gases Components within the fuel-gas and oxidant-paths are resistant to: – specified gases – vapour of solvents		
11.1.102	Solvents Parts are resistant to: – solvents – vapour of solvents		
11.1.103	Parts in contact with acetylene Are not: – copper and alloys with more than 70 % copper – copper alloy filters or sieves – silver or silver alloys except for brazing – for brazing: – silver less than 43 % and – copper less than 21 % and – solder gap less than 0,3 mm		
11.2	F Cleaning (see Form A.20)		
11.3	F Spillage (see Form A.20)		
11.4	F Overflow (see Form A.20)		
11.5	Liquid leakage (AM 1 only)		
11.5.1	Equipment containing liquid (AM 1 only) F (see Form A.20)		
11.5.2 or 11.5 (AM 2)	Battery electrolyte (AM 1 only) Battery electrolyte Battery electrolyte leakage presents no hazard		

Clause Subclause	Requirement	Result	Comments
11.6 F	Specially protected equipment (see Form A.20)		
11.7 (AM 2)	Fluid pressure and leakage		
11.7.1 (AM 2)	Maximum pressure not exceeded		
11.7.2 (AM 2) F	Leakage and rupture at high pressure (see Form A.21)		
	Test to IEC 60335 (refrigeration only)		
11.7.3 (AM 2) F	Leakage from low-pressure parts (see Form A.21)		
11.7.4 (AM 2)	Overpressure safety device		
	– shall not operate in NORMAL USE and		
	– shall comply with the following:		
	– positioned close to parts intended to be protected		
	– access for inspection, maintenance and repair		
	– adjustment only with TOOL		
	– no discharge to person		
	– no hazard from discharge		
	– sufficient discharge capacity		
	– no shut-off valve between protective device and protected parts		
12	Protection against radiation, including laser sources, and against sonic and ultrasonic pressure		
12.1	General		
12.2	Equipment producing ionizing radiation		
12.2.1 F	Ionizing radiation (see Form A.22)		
12.2.2	Accelerated electrons		
12.3	Ultra-violet radiation		(Conformity test under consideration)
12.4	Micro-wave radiation		(Conformity test and limit of 10 W/m ² are under consideration)

Clause Subclause	Requirement	Result	Comments
12.5	Sonic and ultrasonic pressure		
12.5.1	Sound pressure level (AM 1 only) (see Form A.23)		
F			
or			
(AM 2)	Sound level (see Form A.23)		
F			
	– occasional sound		
12.5.2	F Ultrasonic pressure (see Form A.23)		
12.6	Laser sources (IEC 60825)		
13	Protection against liberated gases, explosion and implosion		
13.1	Poisonous and injurious gases		
	Attach any data/test reports used to demonstrate conformity		
13.2	Explosion and implosion		
13.2.1	Components		
	Components liable to explode:		
	– pressure release device		
	or		
	– the apparatus incorporates OPERATOR protection (see also 7.5)		
	Pressure release device:		
	– discharge without danger		
	– not obstructable		
	– if an explosion could occur and no pressure relief device, protection is incorporated in equipment		
	– for burners designed to prevent flash-back, the flame is extinguished without flash		
	– for other burners simulating flash-back, no parts expelled outside the equipment		

Clause Subclause	Requirement	Result	Comments
13.2.2	Batteries		
	Explosion or fire hazard:		
	– protection incorporated in the equipment		
	or		
	– instructions specify batteries		
	and		
	– single component cannot cause hazard (short circuit and open circuit) (see Form A.24, including circuit diagram)		
	F		
	– warning marking or symbol 14		
	Battery compartment design		
13.3	Polarity reversal test		
	Implosion of high-vacuum devices		
	High vacuum devices:		
	– intrinsically protected and correctly mounted		
	or		
	– ENCLOSURE provides protection:		
13.101	– screen not removable without TOOL		
	– if glass screen, not in contact		
	Flame ignition system		
	Automatic flame ignition:		
	– flame ignited with less than 0,8 l of fuel gas delivered		
	or		
	– ignition procedure stops		
	and		
	– system returns to stand-by		
	Non automatic systems:		
	– not possible to open valves		
	or		
	– manually operated devices:		
	– override of shut-off device		
	– flow rates adjustable		
	– If no ignition, gas flow valve closes		

Clause Subclause	Requirement	Result	Comments
13.102	Non-interchangeability of gas connectors		
	Connectors not interchangeable (fuel gas and oxidant gas)		
13.103	Interchangeable burners		
	Operable only with appropriate gas		
14	Components		
14.1	General		
	Where safety is involved, components meet relevant requirements (see table 3 of this report and figure 5 of IEC 61010-1 – AM 2)		
14.2	Motors		
14.2.1	Motor temperatures		
F	(see Form A.25)		
14.2.2	Series excitation motors		
14.3	Overtemperature protection devices		
F	Devices operating in a SINGLE FAULT CONDITION (see Form A.26)		
	and have/are:		
	– reliable function		
	– RATED to interrupt maximum voltage and current of circuit		
	– RATED for maximum surface temperature of 4.4.4.2		
(AM 2)	– RATED for maximum temperature of 9.2 for parts in contact with flammable liquid		
(AM 2)	– not self-resetting unless protected part cannot function		
14.4	Fuse holders		
	No access to HAZARDOUS LIVE parts		
14.5	Mains voltage selecting devices		
	Accidental change not possible		
14.6	HIGH INTEGRITY components		
	Used in applicable positions (see table 3)		

Clause Subclause	Requirement	Result	Comments
	Conforms with IEC publications		
	Not a single electronic device		
14.7	Mains transformers		
14.7.1	Short-circuit tests		
F	Transformers meet 4.4.4.1 to 4.4.4.3. (see Form A.27)		
14.7.2	Overload tests		
	Transformer:		
	– has overtemperature protection meeting 14.3		
	or		
F	– meets 4.4.4.1 to 4.4.4.3 (see Form A.28)		
14.8 (AM 2)	Overpressure safety devices		
	Meets ISO 4126		
15	Protection by interlocks		
15.1	General		
	Interlocks are designed to remove a hazard before OPERATOR exposed		
	If exception for ≤ 2 s (AM 1 only) :		
	– warning marking		
	and		
	– after 2 s:		
	i) temperatures of easily touched parts meet table 3 of IEC 61010-1		
	ii) moving parts meet 7.2 of IEC 61010-1		
	Position of warning markings (AM 1 only)		
15.2	Prevention of reactivation		
15.3	Reliability		
16 (AM2)	Measuring circuits		
16.1 (AM 2)	Current measuring circuits		
F	(see Form A.29)		
Annex K (AM 2)	Routine tests		
	Manufacturer's declaration		

Summary of SINGLE FAULT CONDITIONS applied (4.4.2)
(see Form A.2 for details of tests)

Subclause	Title	Does not apply	Carried out	Comments
4.4.2.1	PROTECTIVE IMPEDANCE			
4.4.2.2	Protective conductor			
4.4.2.3	Equipment or parts for short-term or intermittent operation			
4.4.2.4	Motors			
4.4.2.5	Capacitors			
4.4.2.6	Mains transformers Attach drawing of mains Tx's showing all protective devices (see Forms A.27 and A.28)			
4.4.2.7	Outputs			
4.4.2.8	Equipment for more than one supply			
4.4.2.9	Cooling – air holes closed – fans stopped – coolant stopped			
4.4.2.10	Heating devices – timer overridden – temperature controller overridden – loss of cooling liquid			
4.4.2.11	Insulation between circuits and parts			
4.4.2.12	Interlocks			
List below all SINGLE FAULT CONDITIONS not covered by 4.4.2.1 to 4.4.2.12				

4.4 Testing in SINGLE FAULT CONDITION – Results

Test subclause	Fault No.	Fault description	Td 4.4.3 (note 1)	How was test terminated Comments	Meets 4.4.4

NOTE 1 – Td = Test duration in h:min:s
Record dielectric strength test on Form A.12 and temperature tests on Form A.18.
Record in the comments column for each test whether carried out during or after SINGLE FAULT CONDITION.

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

Form A.3

5.1.3.c Mains supply

Marked RATING: _____ _____ Phase _____ Hz _____ A _____ W _____ VA						NOTE – Measurements are only required for marked RATINGS
Test No.	Voltage V	Freq. Hz	Current I	Power in W	Power in VA	Comments
General comments: _____ _____ _____						

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

5.3 Durability of markings

Table A.4.1 – Marking method (note)
NOTE – Where applicable include print method, label material, ink or paint type, fixing method, adhesive and surface to which marking is fixed.

Table A.4.2 – Agent
A (specify agent)
B (specify agent)
C Water
D Isopropyl alcohol

Marking location	Marking method (see table A.4.1)
Identification (5.1.2)	
Mains supply (5.1.3)	
Fuses (5.1.4)	
Measuring circuit TERMINALS (5.1.4)	
TERMINALS and operating devices (5.1.6)	
DOUBLE/REINFORCED equipment (5.1.7)	
Battery charging (5.1.8)	
Warning marking (5.2)	

Method (table A.4.1)	Test agent (table A.4.2)	Remains legible Pass/Fail	Label loose Pass/Fail	Curled edges Pass/Fail	Comments

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

Clause 6 – Protection against electric shock – Block diagram of system

POLLUTION DEGREE: _____ INSTALLATION CATEGORY (OVERVOLTAGE CATEGORY): _____

Location or description	Insulation type (note 1)	Maximum working voltage (note 2)	CREEPAGE DISTANCE (note 3)				CLEARANCE (note 3) mm	Test voltage (note 2) V	Comments
			PWB mm	CTI	Other mm	CTI			

NOTE 1 – Type of insulation:
 BI = BASIC INSULATION RI = REINFORCED INSULATION
 DI = DOUBLE INSULATION SI = SUPPLEMENTARY INSULATION
 PI = PROTECTIVE IMPEDANCE

NOTE 2 – Types of voltage
 Peak impulse test voltage (pulse) d.c.
 r.m.s. peak

NOTE 3 – INSTALLATION CATEGORIES (OVERVOLTAGE CATEGORIES) or POLLUTION DEGREES which differ from these should be shown under "Comments".

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

6.1.1 Exceptions**6.2 Determination of ACCESSIBLE parts****List of accessible parts**

Item	Description	Determination method (note 5)	Exception under 6.1.1 (note 4)

NOTE 1 – Test fingers and pins are to be applied without force unless a force is specified (see **6.2.1**)

NOTE 2 – Special consideration should be given to inadequate insulation and high voltage parts (see **6.2 AM 2**)

NOTE 3 – Parts are considered to be ACCESSIBLE if they could be touched in the absence of any covering which is not considered to provide suitable insulation (see note to paragraph 1 of **6.4**).

NOTE 4 – Capacitor test may be required (see Form **A.7**).

NOTE 5 – The determination methods are: visual; rigid test finger; jointed test finger; pin 3 mm diameter; pin 4 mm diameter.

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

11.2 Cleaning
11.3 Spillage
11.4 Overflow

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

[illegible]

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

6.5.1.1 Cross-sectional area bonding conductors

Conductor location	Cross-sectional area mm ²	Result Pass/Fail

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

6.5.1.2 Bonding impedance of plug-connected equipment

ACCESSIBLE part under test	Test current A	Voltage attained after 1 min V	Calculated resistance (maximum allowed 0,1 Ω)	Result Pass/Fail

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

6.5.1.3 Bonding impedance of PERMANENTLY CONNECTED EQUIPMENT

ACCESSIBLE part under test	Test current A	Voltage attained after 1 min (maximum 10 V) V	Result Pass/Fail

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

6.5.1.4 Indirect bonding for measuring and test equipment

ACCESSIBLE part under test	Voltage attained	Time for voltage to drop to allowable levels	Result Pass/Fail
a) Voltage limiting device			
	Voltage applied V	Time for device to trip	
b) Voltage-sensitive tripping device			

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

6.5.3 PROTECTIVE IMPEDANCE

A HIGH INTEGRITY single component		
Component	Location	Comments

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

A combination of components		
Component	Location	Comments

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

A combination of BASIC INSULATION and a current or voltage limiting device		
Component	Location	Comments

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

10.1 Integrity of CLEARANCES and CREEPAGE DISTANCES

NOTE – Refer to Form A.12 for dielectric strength tests following the above tests.

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

6.8 Dielectric strength tests

Location (see Form A.5 and/or fault Form A.2)	Working voltage V	Test voltage r.m.s./peak/d.c.	Result Pass/ Fail	Comments (note)

NOTE – Describe conditions prior to testing:

- A = Humidity preconditioning (6.8.2 and 6.8.3)
- B = ENCLOSURE tests (clause 8)
- C = Resistance to heat of non-metallic ENCLOSURES (10.2)
- D = After single faults (4.4)

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

6.10.2.2 Cord anchorage

Location	Mass kg	Pull N	Result Pass/Fail	Torque Nm	Result Pass/Fail	Comment

General comments:

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

9.1 General
Annex G – Test details

Between parts or circuits		CREEPAGE DISTANCE mm	CLEARANCE mm	Working voltage V	Test voltage r.m.s./peak/d.c. V	Result Pass/ Fail	Comments

Tested by:_____Date:_____Test equipment No. (Table 2)_____

9.1 General
Annex F – Test details

F.2.1 Limited circuits

Circuit/ Location	Open circuit voltage r.m.s./d.c. V	Energy limitation					Comments
		Maximum current A	Maximum available power VA	Overload protection	Limited circuit Yes/No	Test to 4.4.3	

General comments:

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

F.2.2 Unlimited circuit

Location/ Circuit	Operator controlled switch	Overcurrent protection	Overtemperature protection	Comments

General comments:

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

9.1 General**Annex F – Test details (continued)****F.4.2 Constructional details****F.4.3 ENCLOSURES**

Clause	Requirement	Result Pass/Fail	Comments
F.4.2.1	Connectors comply with IEC standards		
F.4.2.2	Printing wiring boards are flame RATED FV 0 or FV 1		
F.4.2.3	ENCLOSURE surrounds unlimited circuits		
F.4.2.4	Wires comply with IEC standards		
F.4.3.1	High current devices: – door or – cover and – means to hold door or cover closed or – gap less than limits		
F.4.3.2	Bottom of ENCLOSURES: – no opening or – with table F.1 and figure F.1 or – components placement comply with figure F.2		
F.4.3.3	Baffle or flame BARRIER: – ENCLOSURE made of metal or – ENCLOSURE made of non-metallic material (FV 0 or FV 1)		

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

9.1 General

Annex F – Test details (continued)

F.4.3 ENCLOSURES (F.4.3.3 test to IEC 60707)

Material tested: _____ Generic name: _____ Material manufacturer: _____ Type: _____ Colour: _____ Conditioning details: _____ _____ _____	Overall result Pass/Fail
--	---

	Sample 1	Sample 2	Sample 3
Thickness of specimen mm			
Duration of flaming after first application s			
Duration after flaming plus glowing after second application s			
Specimen burns to holding clamp Yes/No			
Cotton ignited Yes/No			
Sample result Pass/Fail			

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

- 9.2 Temperature tests
- 9.3 Guards
- 9.4 Field wiring TERMINAL boxes

Operating conditions: Frequency: _____ Hz Duration: _____ h _____ min
Voltage: _____ V Test room ambient: _____ °C

Part	t_m °C	t_c °C	t_a °C	Result Pass/Fail	Comments
NOTE 1 – See also 14.1 with reference to component operating conditions. NOTE 2 – t_m = measured temperature t_c = corrected maximum temperature ($t_m + 40$ – test room ambient °C) t_a = maximum permitted temperature.					

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

10.2 Resistance to heat of non-metallic enclosures

Test method used: Non operative treatment []
 Empty ENCLOSURE []
 Operative treatment []

Temperature during tests: _____ °C

ENCLOSURE samples tested were:

Description	Material	Result Pass/Fail	Comments

Dielectric strength test (6.8.4): _____ V _____ r.m.s./peak/d.c.

Comments:

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

[illegible]

NOTE 2 – Use r.m.s., d.c. or peak to indicate the used test voltage.

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

11.7.2 Leakage and rupture at high pressure (AM 2)

Part	Maximum permissible working pressure MPa	Factor (fig. 4 of IEC 61010-1 AM 2)	Test pressure MPa	Leakage test Pass/Fail	Burst test Pass/Fail	Comments

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

11.7.3 Leakage from low-pressure parts (AM 2)

Part	Test pressure MPa	Leakage test Pass/Fail	Comments

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

12.2.1 Ionizing radiation

Locations tested	Measured values μSv/h	Result Pass/Fail	Comments

General comments:

Tested by:_____Date:_____Test equipment No. (Table 2)_____

12.5.1 Sound pressure level (AM 1 only)

12.5.1 Sound level (AM 2 only)

Locations tested	Measured values dBA	Calculated maximum sound pressure level (AM 2)
1 m from the enclosure (AM 1 only)		
At OPERATOR'S normal position and at bystanders' positions (AM 2)		
a)		
b)		
c)		
d)		
e)		
Result Pass/Fail (AM 1 only)		
Comments:		

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

12.5.2 Ultrasonic pressure

Locations tested	Measured values		Comments
	dB	kHz	
At OPERATOR'S normal position			
At 1 m from the ENCLOSURE			
a)			
b)			
c)			
d)			
e)			
NOTE – No limit is specified at present, but a limit of 110 dB above the reference pressure value of 20 µPa is under consideration for applicable frequencies between 20 kHz and 100 kHz.			
Result Pass/Fail			
Comments:			

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

13.2.2 Batteries

Battery load and charging circuit diagram:

Battery type: _____

Battery manufacturer/model/catalogue No.: _____

Battery RATINGS: _____

Reverse polarity instalment test – Result (Pass/Fail): _____

Single component failures	Result	
	Pass/Fail	
Component	Open circuit	Short circuit

Comments:

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

Operating conditions:

Frequency: _____ Hz: _____ Duration: _____ h _____ min

Voltage: _____ V Test room ambient: _____ °C

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

14.3 Overtemperature protection devices

Component	Type (note)	Result Pass/Fail	Comments
NOTE SR = self-resetting (200 times) NSR = non-self-resetting (10 times) NR = non-resetting (1 time).			

General comments:

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

4.4.2.6 Mains transformer

14.7.1 Short-circuit tests (for mains transformers)

Type: _____ Manufacturer: _____				
Tested ☐ in equipment or ☐ on bench				
Optional – Insulation class (IEC 60085) of the lowest RATED winding:				
Winding identification				
Type of protector for winding (note 1)				
Elapsed time				
Current A	Primary			
	Secondary			
Winding temperature °C (note 2)	Primary			
	Secondary			
Tissue paper/cheesecloth OK? (Pass/Fail)				
Voltage tests (note 3)				
Primary to secondary _____ V _____				
Primary to core _____ V _____				
Secondary to secondary _____ V _____				
Secondary to core _____ V _____				
Result (Pass/Fail)				
NOTE 1 – Primary fuse PF / () A Secondary fuse SF / () A Overtemperature protection OP / () °C Impedance protection Z. NOTE 2 – Indicate method of measurement TC = with thermocouple R = resistance method If resistance method is used, record resistance in cold and warm condition under "Comments". NOTE 3 – Record the voltage applied and the type of voltage (r.m.s./d.c./peak) and for results use NB = no breakdown or B = breakdown.				
Comments:				

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

4.4.2.6 Mains transformer**14.7.2 Overload tests (for mains transformers)**

Type: _____ Manufacturer: _____				
Tested ☐ in equipment or ☐ on bench				
Optional – Insulation class (IEC 60085) of the lowest RATED winding:				
Winding identification				
Type of protector for winding (note 1)				
Elapsed time				
Current A	Primary			
	Secondary			
Winding temperature °C (note 2)	Primary			
	Secondary			
Tissue paper/cheesecloth OK? (Pass/Fail)				
Voltage tests (note 3)				
Primary to secondary _____ V _____				
Primary to core _____ V _____				
Secondary to secondary _____ V _____				
Secondary to core _____ V _____				
Result (Pass/Fail)				
NOTE 1 – Primary fuse PF / () A Secondary fuse SF / () A Overtemperature protection OP / () °C Impedance protection Z. NOTE 2 – Indicate method of measurement TC = with thermocouple R = resistance method If resistance method is used, record resistance in cold and warm condition under "Comments". NOTE 3 – Record the voltage applied and the type of voltage (r.m.s./d.c./peak) and for results use NB = no breakdown or B = breakdown.				
Comments:				

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

16.1 Current measuring circuits (AM 2 only)

The test is performed with all types and models of current transformers without internal protection, and which are specified by the manufacturer for use with the equipment.

a) Current transformers

Type/Model	RATED current A	Test current A	Interrupt Yes/No	Result Pass/Fail	Comments

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____

b) Range changing switches

Type/Model	Maximum RATED current of switch	Cycling test Pass/Fail	Comments

Tested by: _____ Date: _____ Test equipment No. (Table 2) _____



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Q6 If you ticked NOT AT ALL in Question 5 the reason is: (tick all that apply)

- standard is out of date ☐
 standard is incomplete ☐
 standard is too academic ☐
 standard is too superficial ☐
 title is misleading ☐
 I made the wrong choice ☐
 other

Q7 Please assess the standard in the following categories, using the numbers:

- (1) unacceptable,
 (2) below average,
 (3) average,
 (4) above average,
 (5) exceptional,
 (6) not applicable

- timeliness.....
 quality of writing.....
 technical contents.....
 logic of arrangement of contents
 tables, charts, graphs, figures.....
 other

Q8 I read/use the: (tick one)

- French text only ☐
 English text only ☐
 both English and French texts ☐

Q9 Please share any comment on any aspect of the IEC that you would like us to know:

.....



ISBN 2-8318-4582-3



ICS 19.080; 71.040.10
